Exercises

Exercise: Linear model with single continuous explanatory variable

1. Either create a new R script (perhaps call it 'linear_model_1') or continue with your previous data exploration R script in your RStudio Project. Again, make sure you include any metadata you feel is appropriate (title, description of task, date of creation etc) and don't forget to comment out your metadata with a `#` at the beginning of the line.

2. Import the data file 'loyn.txt' you used in the previous exercise into R and remind yourself of the data exploration of these data you have already performed (graphical data exploration exercise). The aim of this exercise is to get familiar with fitting a simple linear model with a continuous response variable, bird abundance (`ABUND`) and a single continuous explanatory variable forest area (`AREA`) in R. Ignore the other explanatory variables for now.

3. Create a scatterplot of bird abundance and forest patch area to remind yourself what this relationship looks like. You may have to transform the `AREA` variable to address potential issues that you identified in your previous data exploration. Try to remember which is your response variable (y axis) and which is your explanatory variable (x axis). Now fit an appropriate linear model to describe this relationship using the `lm()` function. Remember to use the `data =` argument. Assign this linear model to an appropriately named object (`loyn_lm` if you imagination fails you!).

4. Display the ANOVA table by using the `anova()` function on your model object. What is the null hypothesis you are testing here? Do you reject or fail to reject this null hypothesis? Explore the ANOVA table and make sure you understand the different components. Refer back to the lectures if you need to remind yourself or ask an instructor to take you through it.

A note on P values and confidence intervals

Before you go on, a short word about how we will report the results of our models for the rest of the course, because it differs a little from what you will most often see in print.

Throughout these exercises you will see P values, and you will see them everywhere in the published literature, so it's important that you can read them. A P value answers one rather narrow question, and it's worth being precise about what that question is.

Imagine that forest patch area had no effect whatsoever on bird abundance, and that we then repeated this whole study over and over again, each time going out and surveying a fresh sample of 67 patches. Purely through the luck of which patches we happened to land on, some of those studies would give us a slope close to zero and others would give us a surprisingly steep one. The P value is the proportion of those imaginary repeat studies that would produce a slope at least as far from zero as the one we actually got. (Notice something rather odd about that. The P value doesn't depend only on the slope we actually measured, it also counts every slope *more* extreme than ours, results which never happened and which we never saw. That is a genuine and long-standing criticism of P values, not a quirk of the way I've explained it here.)

So a small P value is telling you that your data would be an unusual thing to see if there really were no effect. It does not tell you the probability that there is no effect, and it certainly doesn't tell you how big an effect is.

The 0.05 threshold is a convention, not a fact about nature. A result at P = 0.049 and a result at P = 0.051 are practically identical, and treating them as two different kinds of finding is a mistake that you will nevertheless see made regularly in print.

Two consequences that we will hold to throughout this course:

1. **Lead with the estimate and its confidence interval, not with the test.** The interval tells you the size of the effect and how precisely you have measured it, which is what a reader actually needs to know.
2. **A large P value never means 'no difference'.** It means that you have not detected one, which might be because there is no difference, or because your study is too small or too noisy to see it. Look at the confidence interval: if it stretches from 'trivial' to 'enormous', then the honest conclusion is that you can't tell.

None of this means that hypothesis tests are useless, or that you should stop using them. You will meet the threshold based style constantly when reading papers, so learn to read it fluently, and get into the habit of translating it into estimates and intervals in your own head as you go. (if you want to read more about P values and how to misinterpret them see here)

5. Now display the table of parameter (coefficient) estimates using the `summary()` function on your model object. Again, make sure you understand the different components of this output and be sure to ask if in doubt. What are the estimates of the intercept and slope? Write down the word equation of this linear model including your parameter estimates (hint: think y = a + bx).

Your estimates of the intercept and slope are just that, estimates. They come from a single sample of 67 forest patches and a different sample of patches would have given you slightly different values. To get a handle on how precisely each parameter has been estimated, use the `confint()` function on your model object to obtain the 95% confidence intervals for both the intercept and the slope. Now interpret the interval for the slope *in birds*: what range of values for the change in bird abundance per 1 unit increase in `LOGAREA` are these data reasonably consistent with? Keep this on the scale of the transformation, which is the scale the model was fitted on (a 1 unit increase in `LOGAREA` is a tenfold increase in patch area on the original scale, but we will come back to the original scale in Question 13).

6. What is the null hypothesis associated with the intercept? What is the null hypothesis associated with the slope? Do you reject or fail to reject these hypotheses? Finally, have a go at stating these same two findings again, but this time without using the word 'significant' anywhere. Which of your two versions tells a reader more?

7. Looking again at the output from the `summary()` function how much variation in bird abundance is explained by your log transformed `AREA` variable?

8. Now onto a really important part of the model fitting process. Let's check the assumptions of your linear model by creating plots of the residuals from the model. Remember, you can easily create these plots by using the `plot()` function on your model object (`loyn_lm` or whatever you called it). Also remember that if you want to see all plots at once then you should split your plotting device into 2 rows and 2 columns using `par(mfrow = c(2,2))` before you create the plots (Section 4.4).

Can you remember which plot is used to check the assumption of normality of the residuals? What is your assessment of this assumption? Next, check the homogeneity of variance of residuals assumption. Can you see any patterns in the 'residuals versus fitted' or the 'Scale-Location' plots? Is there more or less equal spread of the residuals? Finally, take a look at the leverage and Cook's distance plot to assess whether you have any residuals with high leverage or any influential residuals. What is your assessment? Write a couple of sentences to summarise your assessment of the modelling assumptions as a comment in your R code.

9. Using your word equation from Question 5, how many birds do you predict if `AREA` is 100?

10. Calculate the fitted values from your model using the `predict()` function and store these predicted values in an object called `pred_vals`. Remember, you will first need to create a dataframe object containing the values of log transformed `AREA` you want to make predictions from. Refer back to the model interpretation video if you need a quick reminder of how to do this or ask an instructor to take you through it if you're in any doubt (they'd be happy to take you through it).

11. Now, use the `plot()` function to plot the relationship between bird abundance (`ABUND`) and your log transformed `AREA` variable. Also add some axes labels to aid interpretation of the plot. Once you've created the plot then add the fitted values calculated in Question 10 as a line on the plot (you will need to use the `lines()` function to do this but only after you have created the plot).

12. Now let's recreate the plot you made in Question 11, but this time we'll add the 95% confidence intervals in addition to the fitted values. This question is worth spending some time on: the confidence band around the fitted line is simply the visual counterpart of the confidence intervals you calculated for the parameters back in Question 5, and it shows you at a glance how precisely this relationship has been estimated. Remember, you will need to use the `predict()` function again but this time include the`se.fit = TRUE` argument (store these new values in a new object called `pred_vals_se`). When you use the `se.fit = TRUE` argument with the `predict()` function the returned object will have a slightly different structure compared to when you used it before. Use the `str()` function on the `pred.vals.se` to take a look at the structure. See if you can figure out how to access the fitted values and the standard errors. Once you've got your head around this you can now use the `lines()` function three times to add the fitted values (as before) and also the upper and lower 95% confidence intervals. Don't forget, if you want the 95% confidence intervals then you will need to multiply your standard error values by the critical value of 1.96. If you need a reminder then take a look at the video on confidence intervals on the course MyAberdeen site (in the 'Introductory statistics lectures' folder which you can find in the 'Course Information' learning module).

13. This one is optional, so feel free to skip it if you've had enough (you can always find the R code for this question in the exercise solutions if you want to refer back to it at a later date). This time plot the relationship between bird abundance (`ABUND`) and the original untransformed `AREA` variable. Now back-transform your fitted values (remember you got these with the `predict()` function) to the original scale and add these to the plot as a line. Hint 1: you don't need to reuse the `predict()` function, you just need to back-transform your `my.data$LOGAREA` values. Hint 2: remember if you used a $\log_{10}$ transformation (`log10()`) then you can back-transform using `10^my.data$LOGAREA` and if you used a natural log transformation then use `exp(my.data$LOGAREA)` to back-transform. Comment on the differences between the plot on the transformed (log) scale and the plot on the back-transformed scale in your R script.

14. Time to write up this analysis. Everything you've worked out in this exercise would end up in your thesis as about three sentences, so let's practice writing them. I've given you the skeleton below and all you need to do is fill in the gaps using your output from Questions 5, 7 and 12. Don't worry about matching my wording exactly, there's no single right answer here.

> Bird abundance increased with forest patch area. A linear model of bird abundance against $\log_{10}$ transformed patch area estimated an increase of _________ birds (95% CI: _________ to _________) for every 1 unit increase in $\log_{10}$ area, which is a tenfold increase in patch area (F, = _________, p _________). The model explained _________% of the variation in bird abundance (n = _________ patches).

End of the linear model with single continuous explanatory variable exercise